Authentication & Security Analysis

This page compares authentication methods used in Power BI, highlights the importance of secure user authentication, and explains how organizational accounts and multi-factor authentication (MFA) help protect sensitive business intelligence data.

Authentication Methods

Users authenticate usinng a userna...

First Description

Additional Security Layer

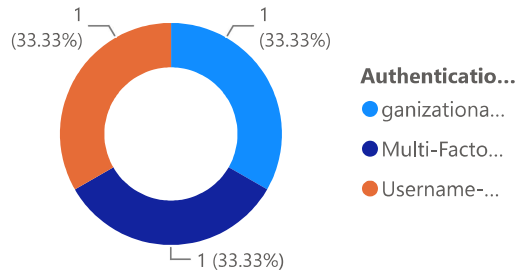
Adds an extra layer of account security

First Security Advantages

Role-Based Access Control Matrix

Role	View	Edit	Share	Admin
CEO	Yes	No	No	No
Finance MAnager	Yes	No	No	No
Regional Manager	Yes	No	No	No
Sales Manager	Yes	No	No	No

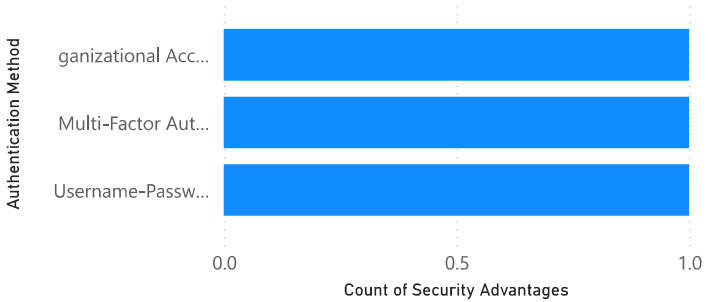
Authentication Methods Overview



Organizational Role and Access Matrix

Role	Reports and Dashboard Access	Information Requirements
CEO	All five reports	Overall sales ,profit, orders a
Finance Manager	KPI Monitoring ,Weekly Profit ,Monthly Executive	Profit, profit margin and fina
Regional Manager	KPI Monitoring ,Monitoring Scenario,daily Sales ,Weekly Profit	Region wise sales ,profit and
Sales Manager	KPI Monitoring ,Daily Sales ,Monthly Executive	Daily sales ,orders, products,

Security Advantages by Authentication Method



Row-Level Security Design

This page illustrates the role-based regional data access strategy using Row-Level Security (RLS). It presents regional access distribution, role assignments, data access coverage, and security restrictions to demonstrate how sensitive business information can be protected from unauthorized access.

Regional Security Role Summary

All Regions
First Assigned Region

All regional d...
First Data Access

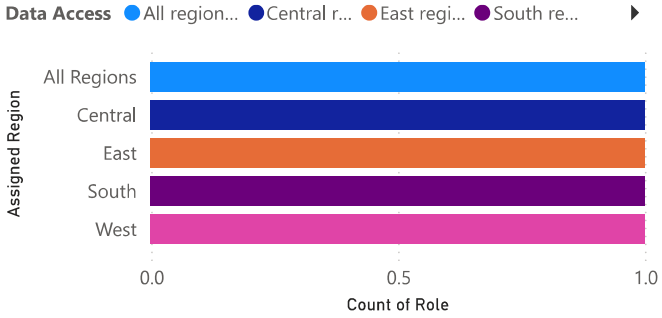
RLS_Design

Role	Assigned Region	Data Access	RLS Filter Condition
CEo	All Regions	All regional data	No regional restriction
Regional Manager-Central	Central	Central region only	Region = Central
Regional Manager-East	East	East region only	Region = East
Regional Manager-South	South	South region only	Region = South
Regional Manager-West	West	West region only	Region=West

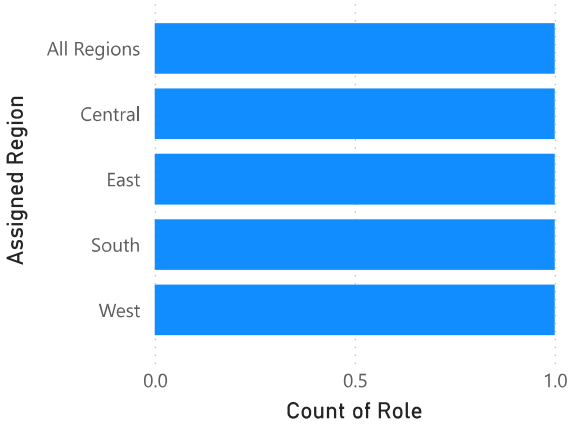
5
Total RLS Roles

4
Regional Restrictions

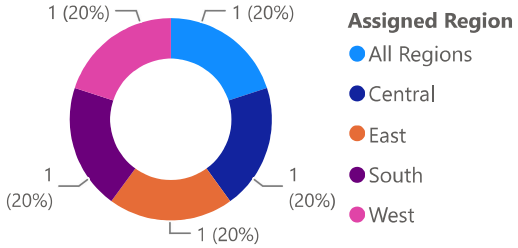
Regional Access Classification



RLS Roles by Assigned Region



Regional Access Distribution



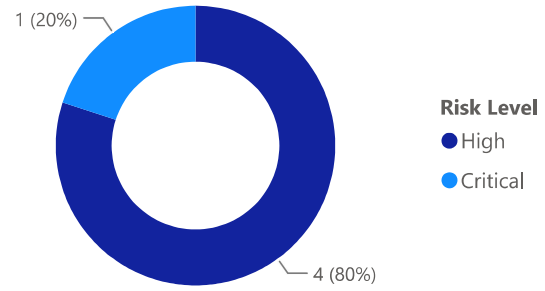
Data Access Coverage



Security Risk Assessment Dashboard

This page evaluates potential BI security threats by analyzing their likelihood, impact, and risk levels. It highlights critical security vulnerabilities, visualizes risk distribution, and presents mitigation strategies and recommended controls to strengthen organizational data protection.

Security Risk Distribution

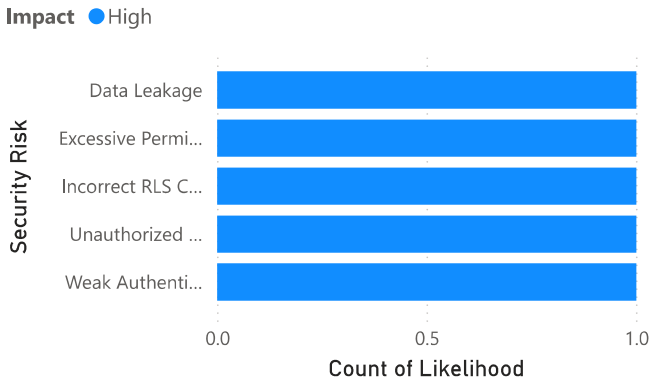


4
High Risk

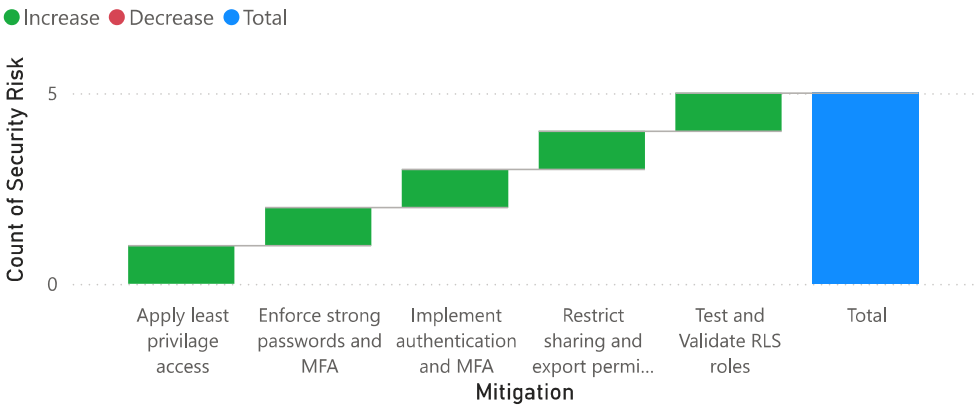
1
Critical Risk

5
Total security Risk

Security Risks by Impact



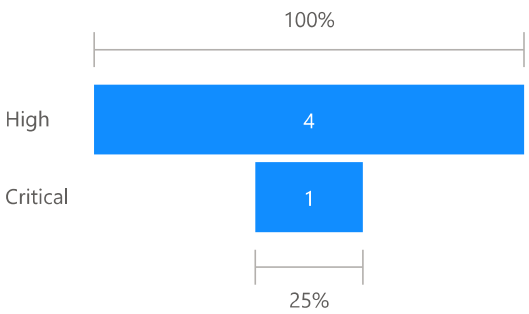
Risk Mitigation Overview



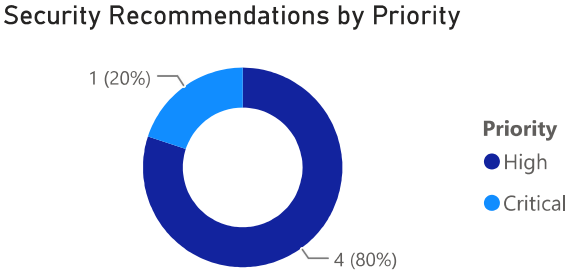
Organizational BI Security Risk Assessment

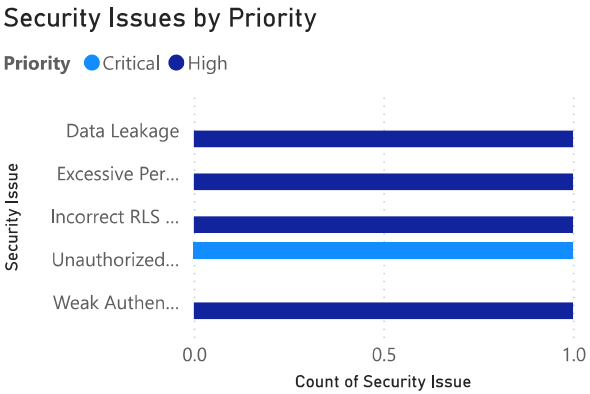
Security Risk	Likelihood	Risk Level	Mitigation	Impact
Data Leakage	Medium	High	Restrict sharing and export permissions	High
Excessive Permissions	Medium	High	Apply least privilage access	High
Incorrect RLS Configuration	Medium	High	Test and Validate RLS roles	High
Unauthorized Access	High	Critical	Implement authentication and MFA	High
Weak Authentication	Medium	High	Enforce strong passwords and MFA	High
Total				

Authentication & Security Recommendations



This page compares authentication methods, highlights the importance of secure user verification, and presents recommended security controls with their priorities and expected outcomes to protect organizational BI reports and sensitive business data.





Security Recommendations Matrix

Recommended Control	Priority	Expected Outcome	Security Issue
Apply Lease Privilage	High	Restricted Unauthorized actions	Excessive Permissions
Enforce Strong Authentication	High	Reduced account security risk	Weak Authentication
Implement MFA	Critical	Stronger user verification	Unauthorized Access
Regularly Test RLS Roles	High	Accurate regional data restrictions	Incorrect RLS Configuration
Restrict Sharing and Exploring	High	Improved data protection	Data Leakage
Total			

3
Authentication Methods

MFA
Additional Security Layer

Authentication Methods Comparison

Authentication Method	Description	Security Advantages
ganizational Account	Users sign in using their organization's Microsoft account	Supports centralized
Multi-Factor Authentication(MFA)	Users verify their identity using an additional authentication factor	Adds an extra layer o
Username-Password	Users authenticate usinng a username and password	Provides basic accou

Security Action Priority Matrix

Security Issue	Critical	High	Total
Data Leakage		Restrict Sharing and Exploring	**Restrict**
Excessive Permissions		Apply Lease Privilage	**Apply L**
Incorrect RLS Configuration		Regularly Test RLS Roles	**Regular**
Unauthorized Access	Implement MFA		**Implem**
Weak Authentication		Enforce Strong Authentication	**Enforce**
Total	**Implement MFA**	**Apply Lease Privilage**	**Apply L**